

# HAONAN GE

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🏠 [Haonan-Ge.github.io](https://Haonan-Ge.github.io)

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## RESEARCH INTERESTS

Large Language Model, Multimodal Reasoning, Video Understanding, Agentic AI, World Model.

## EDUCATION

University of California, Santa Barbara (Advisor: Prof. Yao Qin)

*Ph.D. in Computer Science*

Santa Barbara, CA

*Sep 2026 – Present*

Southeast University

*B.Eng. in Information Engineering; GPA: 3.72/4.0*

Nanjing, China

*Sep 2022 – Jun 2026*

University of California, Irvine

*Exchange in Computer Science; GPA: 4.0/4.0*

Irvine, CA

*Jun 2025 – Sep 2025*

## PUBLICATIONS

\* denotes equal contribution

**What Should a Streaming Video Model Remember?**

Haonan Ge, Yiwei Wang, Hang Wu, Yujun Cai.

*NeurIPS 2026.*

**FrameMind: Frame-Interleaved Video Reasoning via Reinforcement Learning**

Haonan Ge, Yiwei Wang, Kai-Wei Chang, Hang Wu, Yujun Cai.

*NeurIPS 2026 Under Review.*

**MRFD: Multi-Region Fusion Decoding with Self-Consistency for Mitigating Hallucinations in LVLMs**

Haonan Ge, Yiwei Wang, Ming-Hsuan Yang, Yujun Cai.

*Findings of EMNLP 2025.*

**SportR: A Benchmark for Multimodal Large Language Model Reasoning in Sports**

Haotian Xia\*, Haonan Ge\*, Junbo Zou\*, Hyun Woo Choi, Xuebin Zhang, Danny Suradja, Botao Rui, Ethan Tran, Wendy Jin, Zhen Ye, Xiyang Lin, Christopher Lai, Shengjie Zhang, Junwen Miao, Shichao Chen, Rhys Tracy, Vicente Ordonez, Weining Shen, Hanjie Chen.

*ICLR 2026.*

**CamReasoner: Reinforcing Camera Movement Understanding via Structured Spatial Reasoning**

Hang Wu, Yujun Cai, Zehao Li, Haonan Ge, Bowen Sun, Junsong Yuan, Yiwei Wang.

*NeurIPS 2026 Under Review.*

**GUI Agents for Continual Game Generation**

Yixu Huang, Bo Li, Na Li, Zhe Wang, Kaijie Chen, Haonan Ge, Qingyi Si, Yuanzhe Shen, Ruihan Yang, Guangjing Wang, Hongcheng Guo.

*EMNLP 2026.*

**RefineShot: Rethinking Cinematography Understanding with Foundational Skill Evaluation**

Hang Wu, Yujun Cai, Haonan Ge, Hongkai Chen, Ming-Hsuan Yang, Yiwei Wang.

## RESEARCH EXPERIENCE

University of California, Los Angeles & The University of Queensland

*Research Intern (Remote); Supervised by Prof. Yujun Cai & Kai-Wei Chang*

Los Angeles, CA

*May 2025 – Present*

**Project 1: What Should a Streaming Video Model Remember?** (NeurIPS 2026; lead)

- Formulated streaming memory as **budgeted online latent evidence allocation** and proposed **SelectStream**, a fixed-capacity latent-memory graph with surprise-driven adaptive windowing and priority-preserving consolidation for selective writing and retention.
- Designed query-conditioned graph reasoning and calibrated latent evidence injection without frame replay or context growth; achieved **82.67%** on StreamingBench, **67.03%** on OVO-Bench, and **74.4%** average accuracy on offline video benchmarks while keeping latency and GPU memory nearly flat as streams grew.

**Project 2:** *FrameMind: Frame-Interleaved Video Reasoning via Reinforcement Learning* (lead)

- Proposed **Frame-Interleaved CoT (FiCOT)**: a multi-turn loop that alternates textual reasoning with active visual perception, enabling the model to *request* video evidence during reasoning (e.g., target frames or short clips).
- Introduced **Dynamic-Resolution Frame Sampling (DRFS)** to expose the policy to temporal-spatial trade-offs; developed **DRFS-GRPO** for group-relative optimization from outcome rewards (no frame-level labels).
- Implemented a compact **agent runtime**(prompt templates, tool schema, parsing/verification, 1–3 turn cap) and ran controlled ablations on short/long-video benchmarks (e.g., MVBench/MLVU/VideoMME).
- Demonstrated strong gains on long-video benchmarks (e.g., MLVU, VideoMME) with improved efficiency and interpretability.

**Project 3:** *CamReasoner: Reinforcing Camera Movement Understanding via Structured Spatial Reasoning*

- Co-developed the **Observation-Thinking-Answer (O-T-A)** paradigm, reformulating camera movement recognition as structured ego-motion inference grounded in explicit spatio-temporal observations.
- Built a large-scale inference trajectory suite with **18k SFT reasoning chains** and **38k RL samples**; implemented a two-stage SFT and **EMA-GRPO** pipeline with format and accuracy rewards for logical alignment.
- Evaluated CamReasoner-7B across camera movement benchmarks, improving Qwen2.5-VL-7B from 73.8% to **78.4%** on binary classification and from 60.9% to **74.5%** on VQA.

University of California, Irvine & Rice University

Irvine, CA & Houston, TX

Research Intern (On-site); Supervised by Prof. **Weining Shen & Hanjie Chen**

Jun 2025 – Sep 2025

**Project:** *SportR: A Benchmark for Multimodal LLM Reasoning in Sports* (co-led)

- Led **badminton** CoT annotations with progressive QA hierarchy; enforced quality control and consistency.
- Designed experiments for infraction/type/penalty/explanation/tactics and explicit **grounding** tasks.
- Implemented **SFT** pipeline and **GRPO**-based **RL** training (reward: LLM-as-Judge for QA; **GIoU** for grounding).
- Conducted RL ablations and runs, validating post-SFT → RL improvements and cross-modal generalization.
- Benchmarked open-source MLLMs (Qwen-VL, LLaVA-Next, GLM-4.5V, DeepSeek-VL) with unified prompts and judge ensemble.

The University of Queensland

Remote

Research Intern (Remote); Supervised by Prof. **Yujun Cai**

Feb 2025 – May 2025

**Project:** *MRFD: Multi-Region Fusion Decoding with Self-Consistency for Mitigating Hallucinations in LVLMS*

(EMNLP 2025 Findings; **first author**)

- Proposed a **training-free** decoding method that selects salient regions via cross-attention, generates per-region responses, and fuses them using **Jensen–Shannon Divergence**-based reliability weights.
- Designed **region-aware prompts** (CoT-inspired) and a **consistency-weighted fusion** procedure to improve factual grounding and reduce hallucinations without modifying model parameters.
- Built evaluation scripts for POPE/CHAIR/MME-Hallucination; observed consistent gains over strong decoding baselines (precision/F1 and CHAIR scores).

## SERVICES

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**Reviewer:** ICLR 2027, NeurIPS 2026, CVPR 2026, WACV 2026/27, ACM MM 2026, CIKM 2026, ARR 2026

## SKILLS

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**Programming:** Python, C/C++; Matlab; HTML/CSS/JS;  $\text{\LaTeX}$

**ML/Systems:** PyTorch, Ray, FSDP; NumPy/Pandas; Git, Docker

**OS:** Linux (Ubuntu), Windows

**Languages:** Mandarin (Native), English (Fluent)

## HONORS AND AWARDS

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**Outstanding Student Award**

*Southeast University*

Nanjing, China

2025